Grade: Check Plus

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The probability distribution of the amount drawn from an ATM at a convenience store is tabulated below. In this table X = the amount withdrawn by a randomly selected customer and f(x) = P[X = x]. A customer can only withdraw the amounts indicated in the first column of the table below.

x	f(x)
10	0.10
20	0.15
40	0.30
60	0.15
80	0.25
100	0.05

a. What is the probability that a randomly selected customer withdraws at least $60 from the ATM? ✓Qa $0.15 + 0.25 + 0.05 = \boxed{0.45}$

b. Show by calculation that the expected value withdrawn from the ATM is $50 and that the standard deviation of the amount withdrawn from this ATM is $26.27. ✓Qb $10(.1) + 20(.15) + 40(.3) + 60(.15) + 80(.25) + 100(.05) = 1 + 3 + 12 + 9 + 20 + 5 = 50$ $E(x) = 50$

c. The manager of this convenience score tells you that an expected value of $50 is not correct because it is impossible to withdraw $50 from the ATM. How would you answer the manager? ✓Qc It is possible, if the average of 2 transactions is $50, i.e. one removes $40 and another removes $60, for $50 average.

d. What is the probability that the amount withdrawn from the ATM will fall within one standard deviation of the mean? ✓Qd If the mean is $50, and SD is $26.27, then $50 - $26.27 = $23.73 to $76.27. $P(40) + P(60) = .3 + .15 = 0.45$ or $\boxed{45\%}$

e. What is the probability that the amount withdrawn from the ATM will fall within two standard deviations of the mean?

$10^2(.1) + 20^2(.15) + 40^2(.3) + 60^2(.15) + 80^2(.25) + 100^2(.05) = 10 + 60 + 480 + 540 + 1600 + 500 = 3190$

$3190 - 50^2 = 3190 - 2500 = 690$, $\sqrt{690} = \$26.27$

$50 +/- 2(26.27) = 50 +/- 52.54$

$-2.54 \le x \le 102.54$ ✓Qe

10, 20, 40, 60, 80, 100

1.00, 100%